

Contact

Email

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Github

<https://github.com/afxalz>

LinkedIn

<https://www.linkedin.com/in/afxale>

Technical Skills

- Advanced:
 - C++ (Eigen, OpenCV)
 - Python (NumPy, Scikit-learn)
 - ROS1 and ROS2
 - Git, Linux, CMake
 - Docker Compose
- Intermediate:
 - Docker, Singularity
 - Gazebo, CoppeliaSim
 - C, Bash scripting
- Beginner:
 - Javascript, HTML, Flask
 - Matlab

Education

2019

B.Tech. in Electrical Engg.

Indian Institute of Technology (IIT)
Guwahati

2022

M.Sc. in Computer Science

Czech Technical University in Prague

Afzal Ahmad

Doctoral Student

First year doctoral student working on multi-robot task planning for long-horizon tasks. I like to take my robots to the field.

Experience

R&D Engineer, Fly4Future, Prague, Czech Republic

2023 - 2025

- Designed and implemented **mission planning** systems using **State Machines** and **Behavior Trees** for multi-robot systems, including drones and ground robots, across various applications.
- Developed a comprehensive motion planning pipeline leveraging **Nav2 (ROS2)** and **PX4 autopilot**, ensuring efficient navigation for both drones and ground robots.
- Created a **scalable and distributed mission intelligence** framework capable of handling a large number of robots and human operators. Successfully field-tested this system with a **group of 10 drones**.
- Designed and implemented **integration testing** pipelines using **ROS2** and **Gazebo** for both Software-In-The-Loop (SITL) and Hardware-In-The-Loop (HITL) analysis.
- Utilized **Docker Compose** and **Bash scripting** to automate the SITL process, resulting in a significant **reduction of testing time by 70%**.
- Successfully deployed the mission execution software stack on remote hardware using **Docker** and **Singularity** containers, integrated with a **CI/CD** pipeline for streamlined deployment and updates.

Graduate Research Fellow, Multi-Robot Systems Lab, Prague, Czech Republic

2020 - 2022

- Worked comprehensively on the **full autonomy stack** for drones, including **SLAM (2D Lidar, Ouster)**, **Estimation (Kalman Filters, Particle Filters)**, **Motion Planning (A*, RRT*)** and **Control (PID, MPC, Potential Fields)** in both simulated and field experiments.
- Played a key role in developing and maintaining the **open-source MRS-UAV system**, designed to facilitate **rapid prototyping** and **large-scale deployment** of multi-robot systems, capable of handling **up to 30 drones**.

Aerial Swarm

published to [ICRA 21](#)

[paper](#)

- Designed one of the **first drone swarms in the world**, capable of flying in **GPS-denied** environments. Notably, these drones could operate autonomously **without any communication** to other drones or an external ground station. This groundbreaking work received **international acclaim** and was featured in the **prestigious IEEE Spectrum magazine**.
- Developed a **novel Potential Function-based motion planning** algorithm that successfully navigated a group of four drones through an unknown forest. This achievement demonstrated significant advancements in autonomous navigation capabilities.

Volunteering

- Organizer [2022 IEEE RAS Summer School on Multi-Robot Systems](#) (180 international researchers)
- Head [4i Lab, IIT Guwahati, India](#) (25 students)

Courses

- Deep Learning
- AI in Robotics
- Statistical Machine Learning
- Combinatorial Optimization
- Probability and Random Processes
- Computational Game Theory
- Advanced Control Systems

Leader-guided Navigation

paper

published in [Bioinspiration & Biomimetics Journal](#)

- Formulated a **novel algorithm** for selecting a reliable leader within a drone swarm based on the **similarity and cohesion between the trajectories** of the drones.
- Extended the Aerial Swarm algorithm by incorporating observed trajectories which resulted in **2x faster flights** without breaking the swarm group cohesion.
- Developed a **SITL pipeline** for repeated experiments with multiple robots inside **Gazebo** simulation using **ROS1** and **Bash scripting**. The system reduced the **testing and analysis time by 60%** by automatically collecting and plotting data from **complex experiments**.

Research Assistant, Multi-Robot Systems Lab, Prague, Czech Republic

2019 - 2020

- Designed a motion planning algorithm using **Rapidly-Exploring Random Trees (RRT)** to explore and navigate through an unknown region using **two drones, simultaneously**.
- Developed a **cooperative exploration** algorithm to carry a load (large wood plank) using multiple drones.
- Designed custom worlds in **Gazebo** and implemented the system in **ROS1**. Performed rigorous analysis of travel time and degree of cooperation in different simulated environments.

Awards

2024 Diversity Scholar [Robotics: Science and Systems](#)

One of **27 international scholars (5% acceptance rate)** to attend the Robotics: Science and Systems Conference (major international conference for robotics).

2022 Cena Wernera von Siemens award

Top 6 (7% acceptance rate) masters thesis for potential impact on the new age of industrial revolution.

2018 Shastri Research Student Fellowship

One of **18 Indian students (10% acceptance rate)** to pursue a research internship at Polytechnique Montreal, Canada.